Assignment 1

SI 315

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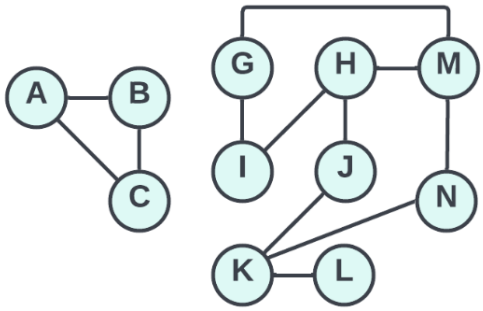
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1 Network 1



1.A) Longest Cycles

Problem

Identify the longest cycles in the network.

The cycle $I \to G \to M \to N \to K \to J \to H \to I$ has 7 nodes and is the only longest cycle.

1.B) Connectedness

Problem

Is the network connected? Explain.

The network is not connected because there are nodes that cannot be reached from other nodes (e.g. $B \to I$).

1.C) Connected Components

Problem

How many connected components does the network have? Which nodes are in each component?

The network has two connected components:

- Component 1: A, B, C
- Component 2: G, M, N, K, L, J, H, I

1.D) Size of Connected Components

Problem

How many nodes are in the largest component of the network?

The largest component is Component 2, which has 8 nodes: G, M, N, K, L, J, H, I.

1.E) Breadth-First Search

Problem

Find the distance from node H to every other node in the network using breadth-first search. Draw a diagram of the breadth-first search procedure similar to the one shown in class, where each new layer of nodes is identified and circled. Remember, the BFS diagram should have all of the nodes in it!

The breadth-first search layers starting at H are:

- Distance 0: H
- Distance 1: I, J, M
- Distance 2: G, K, N
- Distance 3: L
- Unreachable from H: A, B, C

1.F) Add One Edge

Problem

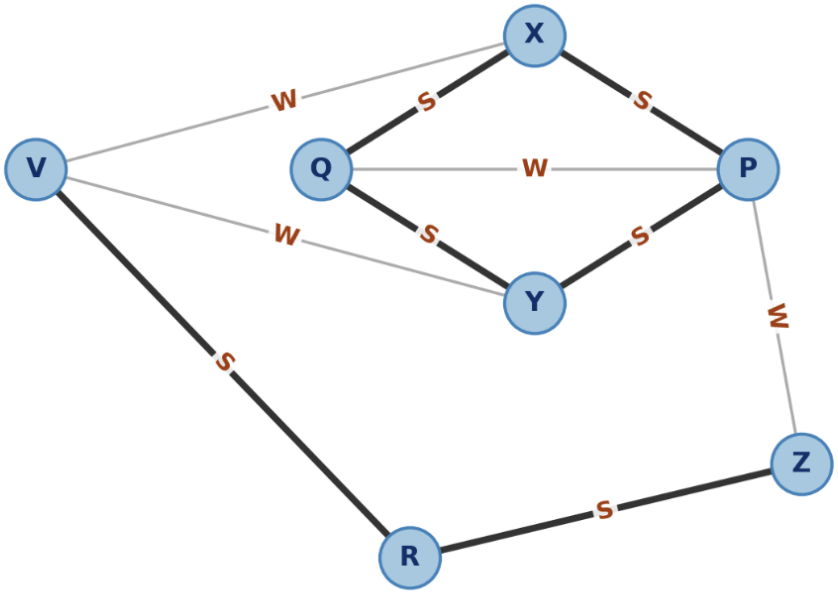
What is the longest cycle you can make by adding one edge to the network? Please specify what edge you would add, how long the longest resulting cycle is, and the cycle itself (i.e., the sequence of nodes). You do not have to explain how you found the edge and the cycle.

The longest cycle can be made by adding the edge $C \to G$. The resulting longest cycle is $I \to G \to C \to A \to B \to C \to G \to M \to N \to K \to J \to H \to I$, which has 11 nodes.

2 Network 2

Problem

In the network below, the ties (P, X), (P, Y), (Q, X), (Q, Y), (R, V), and (R, Z) are labeled strong (S), while all others are labeled weak (W).



Problem

Identify all nodes that violate the Strong Triadic Closure Property. For each one, name its strong ties and the missing edge.